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BWSI Medlytics

**Week 1 Group 8:
Hyperthyroidism-Class
Machine Learning Analysis
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Hyperthyroidism-Class
Machine Learning**

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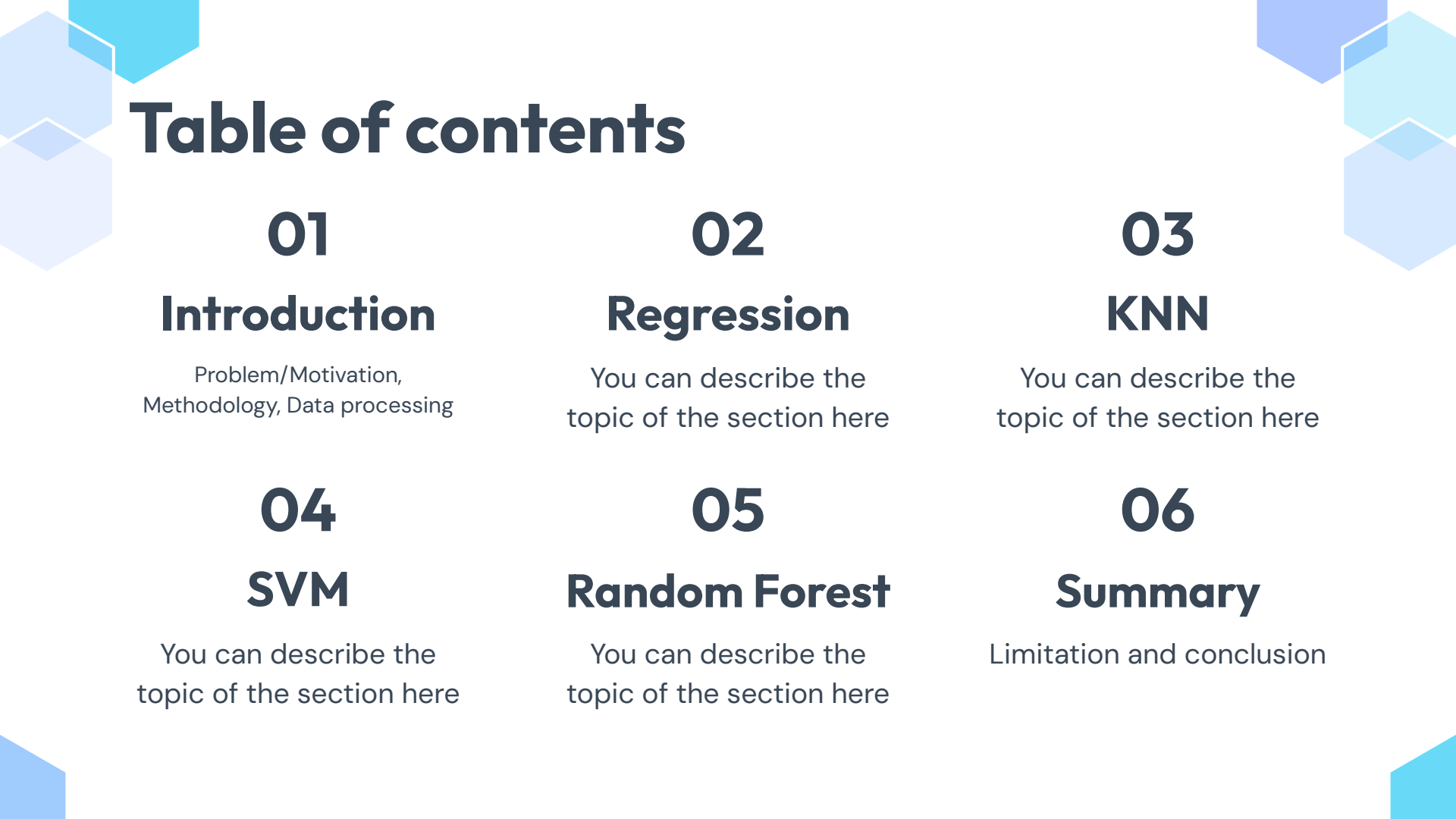


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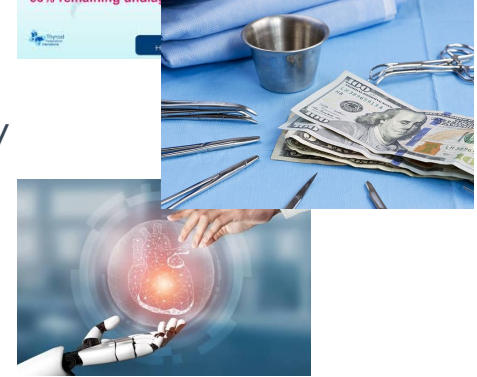
Summary

Limitation and conclusion

Problem/Motivation

- Global Impact: Affects hundreds of millions worldwide, but frequently goes undiagnosed because symptoms like fatigue are easily missed.
- The Cost Barrier: Accurate diagnosis requires tracking up to 30 different clinical attributes, making traditional blood testing highly expensive.
- The ML Solution: Training machine learning models to accurately predict hypothyroidism offers a faster, cheaper, and more accessible screening tool.

Globally,
200
million people
are affected by
thyroid issues, with more than
50% remaining undiagnosed

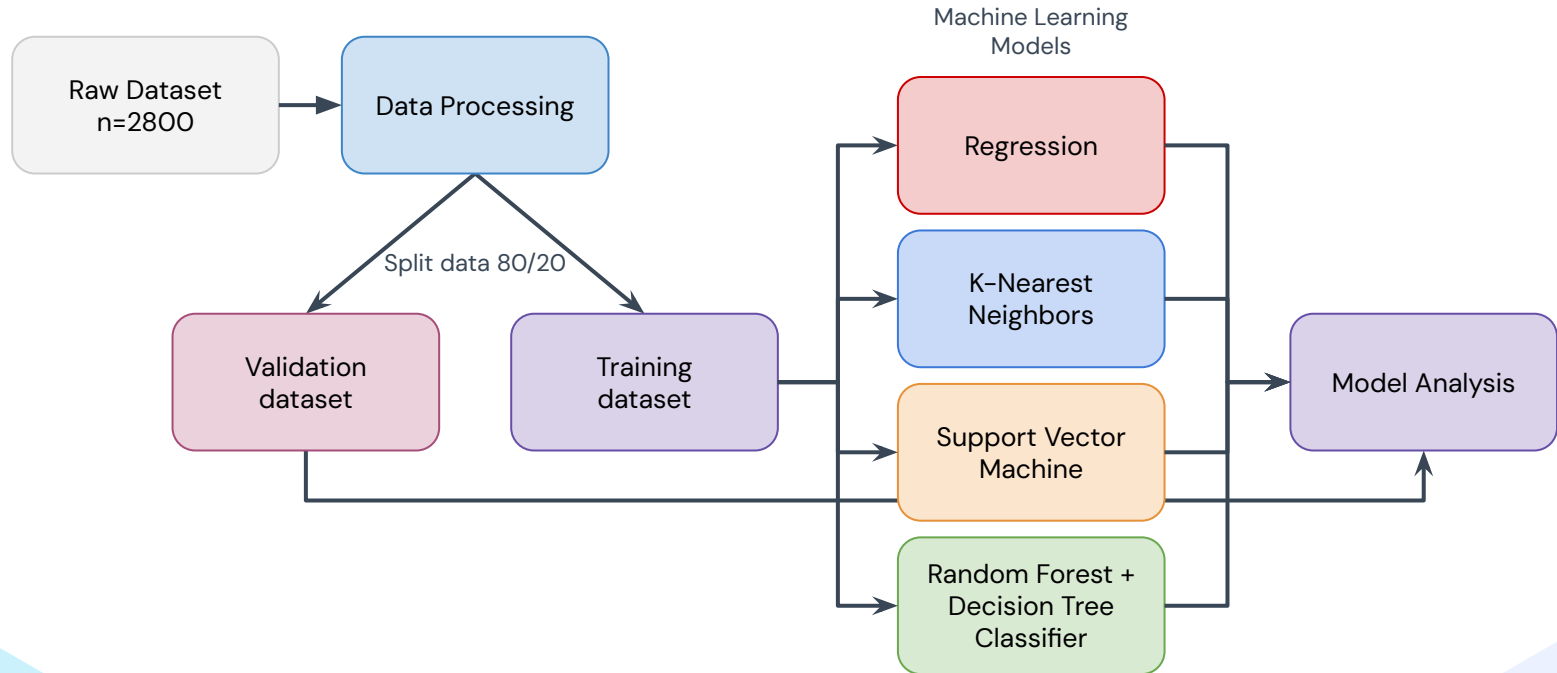


Research Question

How effectively can different machine learning models predict hypothyroidism, and which models most efficiently identify the key clinical features?

Methodology

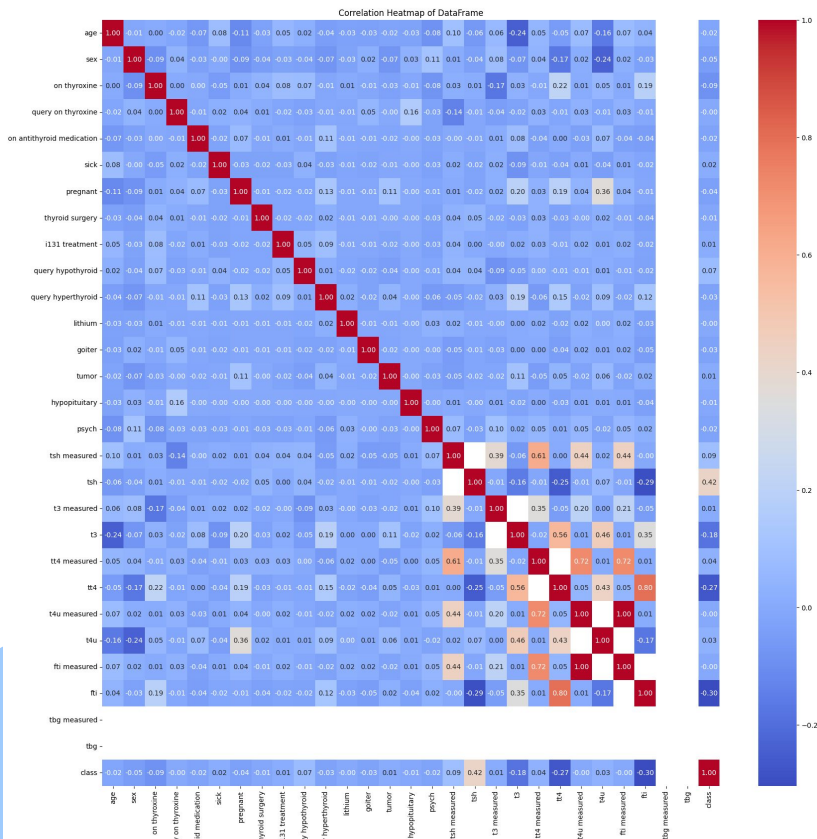
What specific features have the biggest impact on the class for hypothyroidism



Processing the data

- Converted all column names to lowercase for consistency
- Replaced '?' values with NaN
- converted the target labels into binary values
 - negative: 0
 - any hypothyroids: 1
- Converted categorical values such as male/female and true/false into 1/0
- Dropped any duplicates
- Dropped tbj
- Numerical columns:
 - values were replaced with the median
- Data was divided into 8/2 for training and validation

Correlation Map



Zero-rule algorithm

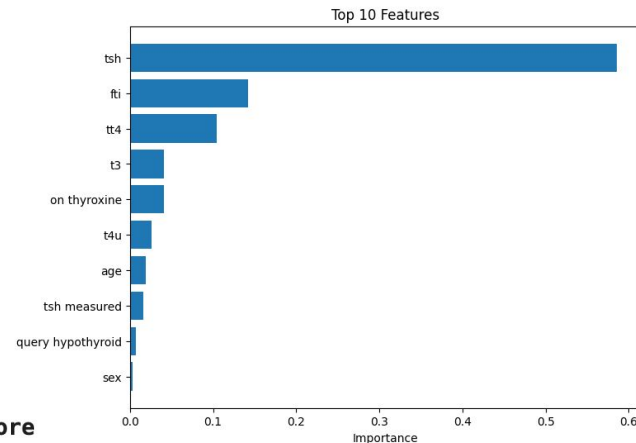
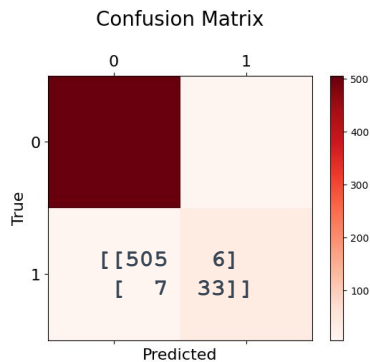
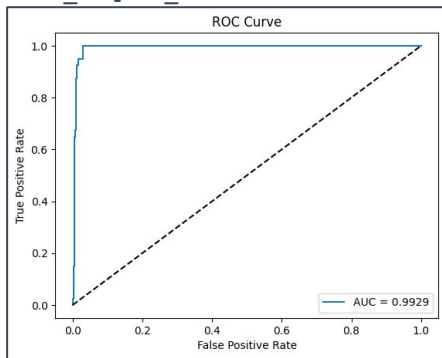
```
def zero_rule_algorithm_classification(train,test):
    majority_class = train['class'].mode()[0]
    return np.full(shape=len(test), fill_value=majority_class)
```

```
y_pred = zero_rule_algorithm_classification(train_df, X_test)
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy*100)
```

Zero-rule accuracy : 92.68%

Random Forest (RF)

```
estimators = 100,  
max_depth = 5,  
min_samples_leaf = 5
```



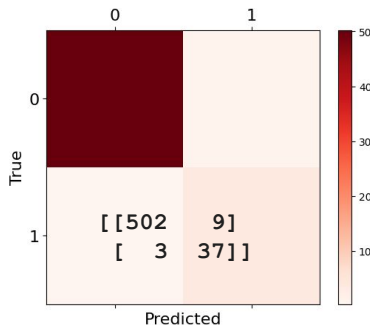
Metric Score

0 Precision 0.804

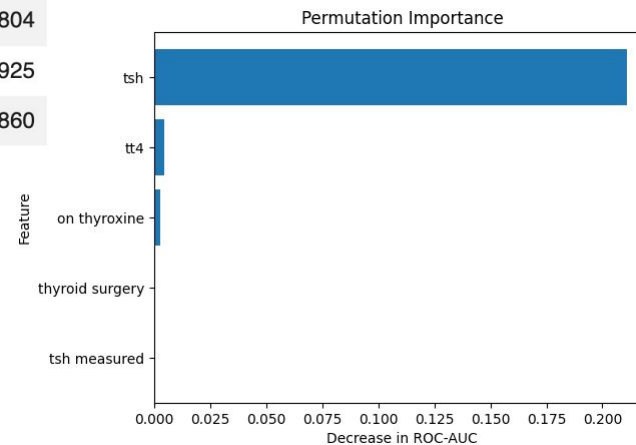
1 Recall 0.925

2 F1 score 0.860

Hyperparameter-tuned Confusion Matrix



Accuracy = 0.9764

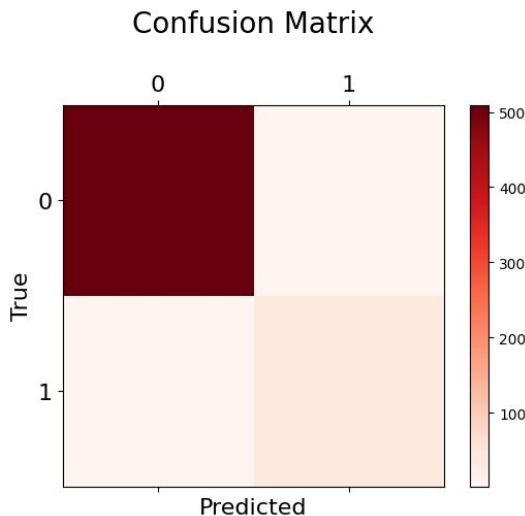
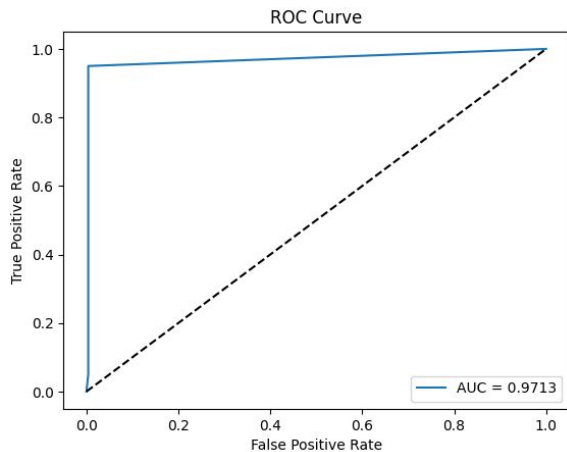


```
estimators = 10,  
max_depth = 30,  
min_samples_leaf = 5
```

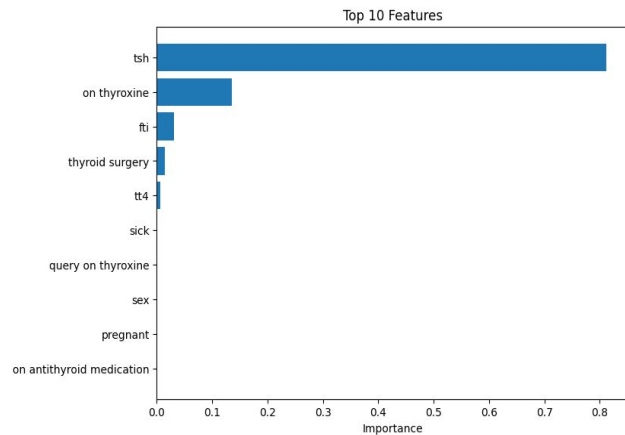
Decision Tree Classifier

Training Accuracy: 0.999

Validation Accuracy: 0.998

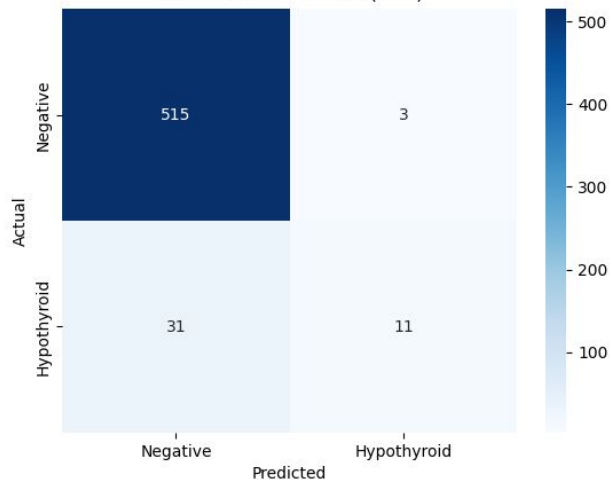


```
[[509  2]
 [  2 38]]
```



K-Nearest Neighbors (KNN)

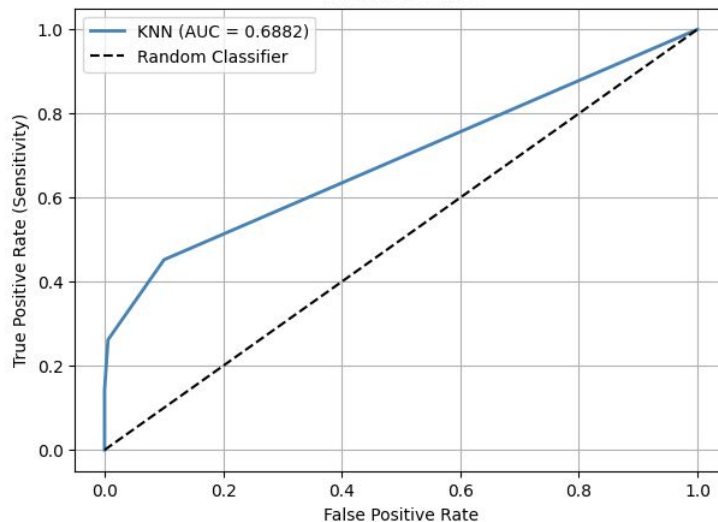
KNN Confusion Matrix (k=3)



Sensitivity: 0.2619
Specificity: 0.9942
Accuracy: 0.9393
AUROC: 0.6882

	precision	recall	f1-score	support
Negative	0.94	0.99	0.97	518
Hypothyroid	0.79	0.26	0.39	42
accuracy			0.94	560
macro avg	0.86	0.63	0.68	560
weighted avg	0.93	0.94	0.92	560

KNN ROC Curve



ROC AUC Score: 0.6882

Training Accuracy: 95.46%

Testing Accuracy: 94.74%

Feature Score

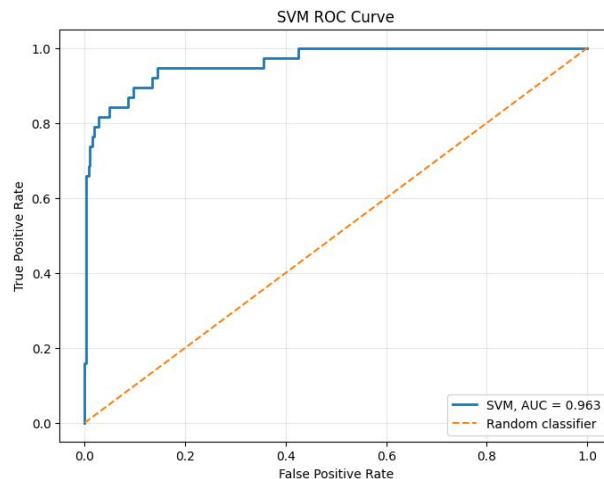
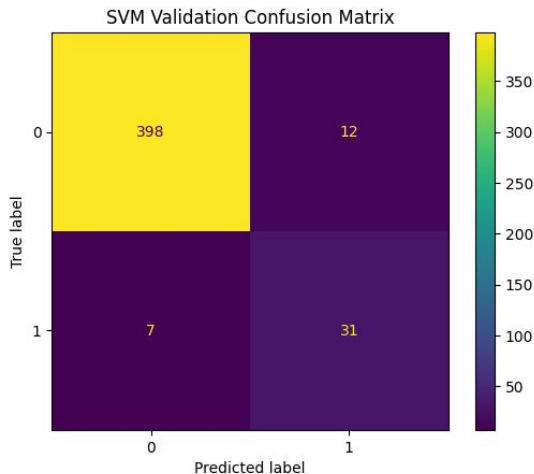
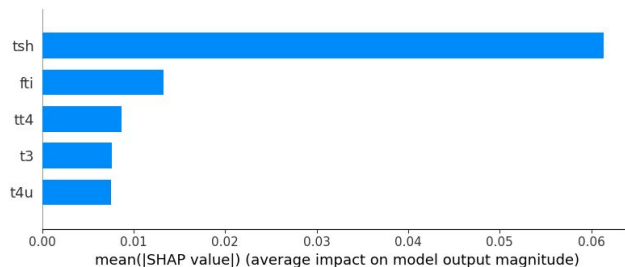
	Feature	Score
17	tsh	373.590703
25	fti	140.310489
21	tt4	114.989715
19	t3	55.285085
2	on thyroxine	18.067219
16	tsh measured	14.890919
9	query hypothyroid	3.955341
1	sex	2.969805
20	tt4 measured	2.687060
6	pregnant	2.547355
23	t4u	1.354567
12	goiter	1.218279
18	t3 measured	0.997887
10	query hyperthyroid	0.745068
7	thyroid surgery	0.744882

Support Vector Machine (SVM)

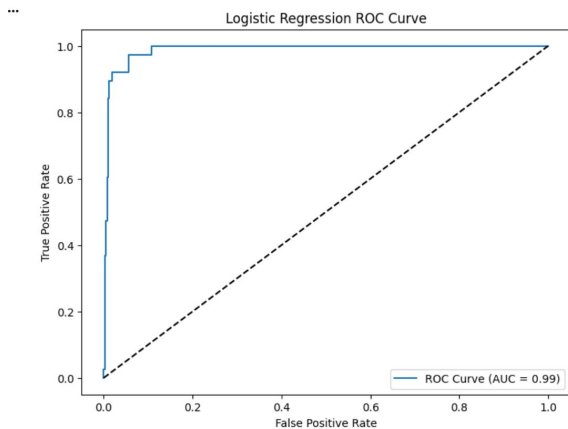
ROC AUC Score: 0.9625

Training Accuracy: 96.32%

Validation Accuracy: 95.76%



Regression

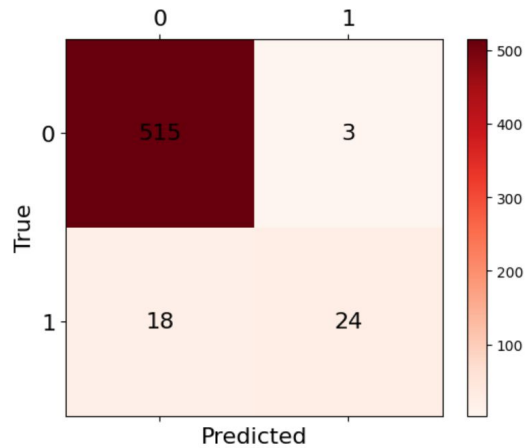


ROC AUC Score: 0.99

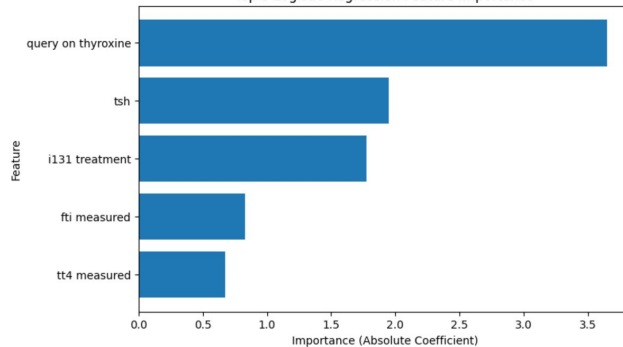
Training Accuracy: 96.71%

Validation Accuracy: 96.25%

Logistic Regression Confusion Matrix



Top 5 Logistic Regression Feature Importance



Limitations

- Treated all non negatives as positive
- Dropped one column, imputation of mean for missing values (may obscure rare patterns)
- Relatively small data set (may not encapsulate rare trends)
- Possibility of overfitting
- Used median scores for certain features





Top Features

TSH – Thyroid Stimulating Hormone, hormone released by the pituitary gland

FTI – Free Thyroxine index, estimate of the amount of biologically available free T4

TT4 – Total Thyroxine

T3 – Triiodothyronine, active thyroid hormone that is more powerful than T4

Query on thyroxine – whether patient had suspicion related to thyroxine

I131 treatment – radioactive iodine treatment, a therapy that destroys part of the thyroid gland

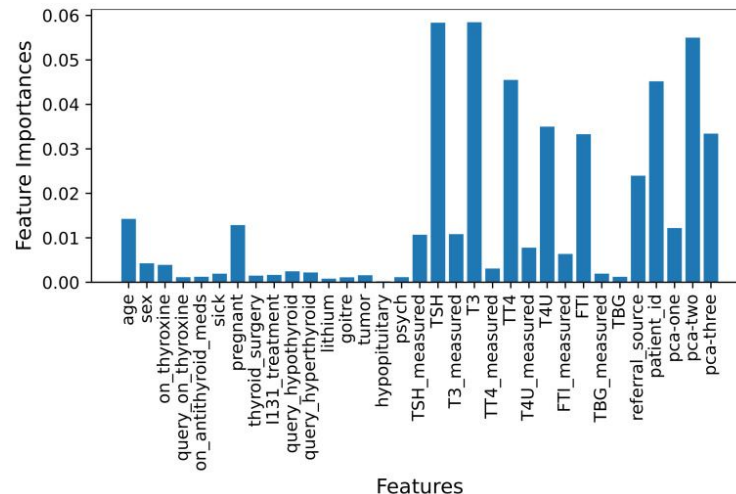


Discussion and Conclusion

Table 13.

Performance of models using MLFS feature set.

Model	Accuracy	Precision	Recall	F1 Score
RF	0.99	0.99	0.99	0.99
GBM	0.98	0.98	0.98	0.98
ADA	0.97	0.97	0.97	0.97
LR	0.87	0.88	0.87	0.87
SVM	0.92	0.92	0.92	0.92



Chaganti R, Rustam F, De La Torre Díez I, Mazón JLV, Rodríguez CL, Ashraf I. Thyroid Disease Prediction Using Selective Features and Machine Learning Techniques. Cancers (Basel). 2022 Aug 13;14(16):3914. doi: 10.3390/cancers14163914. PMID: 36010907; PMCID: PMC9405591.